A MORPHISM BETWEEN INTEGRAL SCHEMES OF EQUAL DIMENSION IS EITHER NON DOMINANT OR DOMINANT WITH FINITE FUNCTION FIELD EXTENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The image of an irreducible space is irreducible, and dominance puts the generic point in the image. Topology, labels topology-lemma-image-irreducible-space and topology-lemma-irreducible (proof lines 1828-1830); Morphisms, label morphisms-lemma-dominant-finite-number-irreducible-components (proof line 1834). The first shows the closure of the image is irreducible, and the second lets the generic point of the target be taken in the image when the morphism is dominant.
- Finiteness of a field extension of transcendence degree zero, under the permanence of finite type. Morphisms, labels morphisms-lemma-finite-degree (proof line 1841) and morphisms-lemma-permanence-finite-type (proof line 1843). Turns the equality of the dimension function values at the two generic points into the statement that the extension of function fields has transcendence degree zero, hence is finite.

Source reference. Original lemma 01RM: sources/stacks-chow/tags.txt, source lines 2307–2307. Navigation pointer for \ref{morphisms-lemma-dominant-finite-number-irreducible-components} used inside the counted span (line 1834); the argument that consumes it is retained in the span.

Date: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02NX: sources/stacks-chow/tags.txt, source lines 3496–3496. Navigation pointer for \ref{morphisms-lemma-finite-degree} used inside the counted span (line 1841); the argument that consumes it is retained in the span.

Source reference. Original lemma 01T8: sources/stacks-chow/tags.txt, source lines 2365–2365. Navigation pointer for \ref{morphisms-lemma-permanence-finite-type} used inside the counted span (line 1843); the argument that consumes it is retained in the span.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 1817); the argument that consumes it is retained in the span.

Source reference. Original lemma 0379: sources/stacks-chow/tags.txt, source lines 4192–4192. Navigation pointer for \ref{topology-lemma-image-irreducible-space} used inside the counted span (line 1829); the argument that consumes it is retained in the span.

Source reference. Original lemma 004W: sources/stacks-chow/tags.txt, source lines 188–188. Navigation pointer for \ref{topology-lemma-irreducible} used inside the counted span (line 1830); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X, Y be locally of finite type over S. Let $f : X \to Y$ be a morphism. Assume X, Y integral and $\dim_\delta(X) = \dim_\delta(Y)$. Then either $f(X)$ is contained in a proper closed subscheme of Y, or f is dominant and the extension of function fields $R(X)/R(Y)$ is finite.*

Proof. The closure $\overline{f(X)} \subset Y$ is irreducible as X is irreducible (Topology, Lemmas 0379 and 004W). If $\overline{f(X)} \neq Y$, then we are done. If $\overline{f(X)} = Y$, then f is dominant and by Morphisms, Lemma 01RM we see that the generic point η_Y of Y is in the image of f. Of course this implies that $f(\eta_X) = \eta_Y$, where $\eta_X \in X$ is the generic point of X. Since $\delta(\eta_X) = \delta(\eta_Y)$ we see that $R(Y) = \kappa(\eta_Y) \subset \kappa(\eta_X) = R(X)$ is an extension of transcendence degree 0. Hence $R(Y) \subset R(X)$ is a finite extension by Morphisms, Lemma 02NX (which applies by Morphisms, Lemma 01T8). $\square$